

Vijayadharshini V 250701832-2025-CSEV2**Started on** Wednesday, 12 August 2026, 10:44 PM**State** Finished**Completed on** Wednesday, 12 August 2026, 10:47 PM**Time taken** 3 mins 6 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3
4  int n;
5  scanf("%d",&n);
6  int i=1;
7  int s=1;
8  int count=3;
9  while(s<=n){
10
11     count++;
12     i++;
13     count++;
14     s+=i;
15     count++;
16 }
17 printf("%d", count);
18 return 0;
19 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

Vijayadharshini V 250701832-2025-CSE

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V2**Started on** Wednesday, 12 August 2026, 10:50 PM**State** Finished**Completed on** Wednesday, 12 August 2026, 11:00 PM**Time taken** 9 mins 55 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      int count=0;
5      scanf("%d",&n);
6      if(n==1){
7          count=1;
8      }
9      else{
10         count=5*n+2;
11     }
12     printf("%d",count);
13
14     return 0;
15 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓

	Input	Expected	Got	
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

Vijayadharshini V 250701832-2025-CSEV2**Started on** Friday, 14 August 2026, 4:06 PM**State** Finished**Completed on** Friday, 14 August 2026, 4:10 PM**Time taken** 4 mins 2 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  void factor(int num){
3      int c=0;
4      int i;
5      for(i=1;i<=num;i++){
6          c++;
7          c++;
8          if(num%i==0){
9              c++;
10         }
11     }
12     c++;
13     printf("%d",c);
14
15 }
16 int main(){
17     int n;
18     if(scanf("%d", &n)==1){
19         factor(n);
20     }
21     return 0;
22 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

Vijayadharshini V 250701832-2025-CSEV2**Started on** Friday, 14 August 2026, 7:51 PM**State** Finished**Completed on** Friday, 14 August 2026, 7:58 PM**Time taken** 6 mins 42 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n,c=0;
4      scanf("%d",&n);
5      c++;
6      for(int i=n/2; ;i++){
7          c++;
8          if(i>=n){
9              break;
10         }
11         for(int j=1; ;j=2*j){
12             c++;
13             if(j>=n){
14                 break;
15             }
16             for(int k=1; ;k=k*2){
17                 c++;
18                 if(k>=n){
19                     break;
20                 }
21                 c++;
22             }
23         }
24     }
25 }
26 printf("%d", c);
27 return 0;
28 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

Vijayadharshini V 250701832-2025-CSEV2**Started on** Friday, 14 August 2026, 7:59 PM**State** Finished**Completed on** Friday, 14 August 2026, 8:03 PM**Time taken** 3 mins 6 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n,rev=0,rem;
4      int c=0;
5      scanf("%d",&n);
6      c++;
7      while(n!=0){
8          c++;
9          rem=n%10;
10         c++;
11         rev=rev*10+rem;
12         c++;
13         n/=10;
14         c++;
15     }
16     c++;
17     c++;
18     printf("%d", c);
19     return 0;
20 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)